Distributed Multi-Variant Model Placement in Edge Computing: A Game-Theoretic Framework*

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Abstract. While edge intelligence has emerged as a promising paradigm to support AI applications, its potential is severely bottlenecked by limited edge server capacities. To satisfy heterogeneous user demands, multiple application service providers (ASPs) are compelled to deploy diverse model variants, which further exacerbates this resource scarcity. Although recent studies have explored model placement and variant selection, the strategic competition among ASPs managing multi-variant portfolios under shared edge capacities remains largely unexplored. In this paper, we investigate the model variants placement problem under limited edge capacities. By incorporating the edge service provider's (ESP) optimal computing resource allocation into ASPs' operational cost, we formulate their strategic placement interaction as a Generalized Nash Equilibrium Problem (GNEP). This problem is intrinsically challenging due to the combinatorial binary placement decisions, the strategy-space coupling induced by the shared storage capacities, and the congestion externalities arising from computing resource contention. To address these challenges, we employ an exact penalty method to decouple the capacity constraints, and rigorously prove that the reformulated penalized game constitutes an exact potential game. This property enables us to design an efficient distributed algorithm with guaranteed finite-step convergence to a pure-strategy Nash Equilibrium (NE). Furthermore, we derive an upper bound for the Price of Anarchy (PoA) under constrained capacity, which gracefully reduces to $(3+\sqrt{5})/2$ in the unconstrained scenario. Simulations demonstrate that the distributed algorithm converges rapidly and significantly outperforms baseline methods in minimizing the social cost. Moreover, empirical evaluations confirm that the actual efficiency loss remains well below our derived theoretical worst-case bounds.

Keywords: Edge computing · Model variants · Game theory

1 Introduction

By bringing computation in close proximity to end users, edge computing enables low-latency model inference, which is crucial for emerging applications such as autonomous driving, smart surveillance, and smart cities [1–3].

* This is the full version of the paper, including the online appendix.

However, across different application scenarios, user demands for latency and accuracy vary significantly, making a single monolithic model inadequate. As a result, application service providers (ASPs) need to deploy diverse model variants, ranging from heavyweight models for tasks requiring high accuracy to lightweight versions for rapid inference. While such multi-variant provisioning ensures service flexibility, it encounters a new challenge due to the limited computing and storage capacities of edge servers. Consequently, when multiple ASPs simultaneously deploy their variant portfolios, severe resource contention is inevitably triggered.

Prior research has explored both model selection and placement to serve heterogeneous latency–accuracy demands under scarce resources. On the model selection side, Zhao et al. [4] propose an online framework jointly optimizing model selection and resource provisioning to balance inference accuracy, latency, and cost. Huang et al. [5] further incorporate deployment locations into this joint optimization under limited edge storage and computation. On the model placement side, Zhang et al. [6] optimize AI service placement and request routing to balance service quality and end-to-end latency. Additionally, Tütüncüoğlu et al. [7] model the interaction between a single operator and end users as a Stackelberg game for joint caching and resource allocation. However, these studies assume a single decision maker, ignoring the competition among multiple independent ASPs that share limited edge capacities.

In this paper, we propose a game-theoretic framework for the competitive placement of heterogeneous model variants. We capture the interactions among self-interested ASPs by analytically embedding the edge service provider's optimal resource allocation into their individual cost structures. Since ASPs contend for shared edge capacities, this interaction naturally constitutes a Generalized Nash Equilibrium Problem (GNEP). However, this GNEP is challenging to solve due to the combinatorial placement decisions, severe computing congestion, and coupled strategy spaces. To address these hurdles, we formulate an exact penalty method to transform the GNEP into a standard Nash Equilibrium Problem (NEP), propose a potential game formulation, and design a distributed algorithm. The main contributions of this paper are summarized as follows:

1. **Novel competitive variants placement framework:** We propose a novel game-theoretic framework for the competitive placement of heterogeneous model variants in edge computing, where self-interested ASPs contend for shared edge capacities to serve heterogeneous demands. By analytically embedding the ESP's optimal resource allocation into the ASPs' cost, the framework naturally captures storage contention, computing congestion, and non-placement penalties, with each ASP strategically minimizing its own operational cost.
2. **GNEP reformulation and exact potential game analysis:** To resolve the coupled strategy spaces induced by the shared storage constraints, we employ an exact penalty method to transform the GNEP into a standard NEP and prove that the penalized game constitutes an exact potential game. We then design a distributed algorithm with guaranteed finite-step conver-

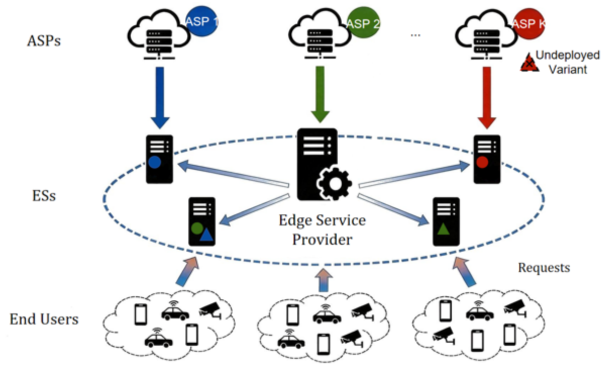

Fig. 1. System model of multi-variant model placement in edge computing.

gence to a NE, and derive a parameterized PoA bound that gracefully reduces to $(3+\sqrt{5})/2$ under abundant storage.

3. **Simulation results:** Extensive simulations validate the finite-step convergence of our proposed distributed algorithm, show that our algorithm consistently achieves the lowest social cost among three benchmarks, and confirm that empirical efficiency losses remain well below the theoretical worst-case bounds.

In the following sections, we first introduce the system model and problem formulation, and then show our game analysis and distributed algorithm design. We finally present the simulation results and conclude the paper.

2 System Model and Problem Formulation

As depicted in Fig. 1, we consider an edge computing system in which an ESP operates a set $\mathcal{S} = \{1, 2, \ldots, S\}$ of heterogeneous edge servers (ESs), each co-located with a base station (BS), to serve a set $\mathcal{K} = \{1, 2, \ldots, K\}$ of ASPs. To satisfy heterogeneous latency and accuracy demands, each ASP $k \in \mathcal{K}$ offers a set $\mathcal{V}_k = \{1, 2, \ldots, V_k\}$ of distinct model variants. Specifically, ASP k handles a request volume $n_{k,v}$ for variant $m_{k,v}$ which requires a computational workload $f_{k,v}$ and a storage requirement $c_{k,v}$. Each ES $s \in \mathcal{S}$ is constrained by a limited storage capacity C_s and computing capacity F_s. Each model variant is placed at most once within the edge service area, and the ESP allocates computing resources in response to the ASPs' model placements. Specifically, we let $x_{s,k,v} \in \{0, 1\}$ be a binary decision variable indicating whether variant $m_{k,v}$ is hosted on ES s, and let $\boldsymbol{x} = \{x_{s,k,v} \mid \forall s \in \mathcal{S}, k \in \mathcal{K}, v \in \mathcal{V}_k\}$ denote the collection of all ASPs' placement decisions. In the following, we first analyze the ESP's optimal resource allocation. Subsequently, we embed this optimal response into the ASPs' cost models to formulate the model variants placement game.

2.1 ESP's Resource Allocation Problem

As the infrastructure manager, the ESP minimizes the total computation latency across all hosted model variants by dynamically slicing its computing resources. Let $\boldsymbol{w}_s = \{w_{s,k,v} \mid \forall k \in \mathcal{K}, v \in \mathcal{V}_k\}$ denote the computing resource allocation strategy on ES s, where $w_{s,k,v} \in [0,1]$ represents the fraction of ES s's total computing capacity F_s assigned to model variant $m_{k,v}$.

When variant $m_{k,v}$ is placed on ES s (i.e., $x_{s,k,v} = 1$), the computation delay on ES s is:

$$T_{s,k,v}(x_{s,k,v}, w_{s,k,v}) = \frac{x_{s,k,v} n_{k,v} f_{k,v}}{w_{s,k,v} F_s}. \tag{1}$$

Given the ASPs' placement decisions $\boldsymbol{x}$, the ESP's resource allocation problem on ES s can be formulated as:

$$\min_{\boldsymbol{w}_s} \quad \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} T_{s,k,v}(x_{s,k,v}, w_{s,k,v}) \tag{2a}$$

$$\text{s.t.} \quad \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} w_{s,k,v} \le 1, \tag{2b}$$

$$0 \le w_{s,k,v} \le 1, \quad \forall k \in \mathcal{K}, v \in \mathcal{V}_k. \tag{2c}$$

Since problem (2a) is strictly convex over a convex feasible region, solving the Karush-Kuhn-Tucker (KKT) conditions yields the unique global optimum.

Lemma 1. *ESP's computing resource allocation problem* (2a) *admits a unique optimal solution $w^*_{s,k,v}$ for each model variant $m_{k,v}$:*

$$w^*_{s,k,v} = \frac{x_{s,k,v}\sqrt{n_{k,v} f_{k,v}}}{\sum_{k'\in\mathcal{K}} \sum_{v'\in\mathcal{V}_{k'}} x_{s,k',v'} \sqrt{n_{k',v'} f_{k',v'}}}. \tag{3}$$

Proof. See Appendix A. ■

2.2 ASPs' Model Variant Placement

Anticipating the ESP's optimal resource allocation $\boldsymbol{w}^* = \{w^*_{s,k,v} \mid \forall s \in \mathcal{S}, k \in \mathcal{K}, v \in \mathcal{V}_k\}$, the ASPs strategically decide their model variant placement policies to minimize their operational expenditure. Specifically, each ASP $k \in \mathcal{K}$ determines a joint placement policy $\boldsymbol{u}_k = (\boldsymbol{x}_k, \boldsymbol{z}_k)$, where $\boldsymbol{z}_k = \{z_{k,v} \mid \forall v \in \mathcal{V}_k\}$ denotes the edge non-placement strategy with $z_{k,v} = 1$ indicating that variant $m_{k,v}$ is not placed at any edge server. We use $\boldsymbol{u}_{-k} = \{\boldsymbol{u}_{k'}\}_{k'\in\mathcal{K}\setminus\{k\}}$ to denote the placement policies of all ASPs other than k.

1) ASP's Operational Cost: An ASP's operational cost is driven by two dominant factors: edge computation cost and the potential loss of edge non-placement (e.g., cloud offloading fees and Service Level Agreement (SLA) penalties [8]). In our model, we omit the transmission delay which is typically small since the BS is in close proximity [9].

Potential Non-Placement Loss: If ASP k does not place variant $m_{k,v}$ at the edge (i.e., $z_{k,v} = 1$), the associated user requests must be either routed to the remote cloud or dropped entirely. To capture this operational overhead, we introduce a consolidated penalty $L_{k,v} > 0$. The aggregated potential loss for ASP k is formulated as:

$$C_k^{Loss}(\boldsymbol{u}_k) = \sum_{v \in \mathcal{V}_k} z_{k,v} L_{k,v}. \tag{4}$$

Computation Cost: The computation cost comes from the local computation delay. Substituting the ESP's optimal response $\boldsymbol{w}^*$ into (1), we set the reformulated local delay for model variant $m_{k,v}$: $T_{s,k,v}(\boldsymbol{x}) = \phi_{s,k,v}(\boldsymbol{x}_k)\varPhi_s(\boldsymbol{x})$, where $\phi_{s,k,v}(\boldsymbol{x}_k) \triangleq x_{s,k,v}\sqrt{n_{k,v} f_{k,v}/F_s}$, $\phi_{s,k}(\boldsymbol{x}_k) \triangleq \sum_{v \in \mathcal{V}_k} \phi_{s,k,v}(\boldsymbol{x}_k)$, and $\varPhi_s(\boldsymbol{x}) \triangleq \sum_{k' \in \mathcal{K}} \phi_{s,k'}(\boldsymbol{x}_{k'})$ capture the per-variant, per-ASP, and aggregate workload indices on ES s, respectively.

Given a latency penalty factor $\beta > 0$, the aggregated computation cost for ASP k is formulated as:

$$C_k^{Comp}(\boldsymbol{u}_k, \boldsymbol{u}_{-k}) = \beta \sum_{v \in \mathcal{V}_k} (1 - z_{k,v}) \sum_{s \in \mathcal{S}} \phi_{s,k,v}(\boldsymbol{x}_k)\varPhi_s(\boldsymbol{x}). \tag{5}$$

Crucially, this formulation isolates the strategic interaction: ASP k's cost is coupled with its rivals' strategies $\boldsymbol{u}_{-k}$ solely through the aggregate function $\varPhi_s(\boldsymbol{x})$. Any unilateral deviation by ASP k alters $\varPhi_s(\boldsymbol{x})$ exclusively via its own independent contributions $\phi_{s,k}(\boldsymbol{x}_k)$, while the rivals' contributions remain strictly fixed.

Consequently, the total operational cost of ASP k is defined as the sum of its computation cost and potential loss:

$$C_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k}) = C_k^{Comp}(\boldsymbol{u}_k, \boldsymbol{u}_{-k}) + C_k^{Loss}(\boldsymbol{u}_k). \tag{6}$$

2) Individual Operational Cost Minimization Problem: As self-interested rational entities, the goal of each ASP $k \in \mathcal{K}$ is to minimize its individual operational cost by optimizing its placement policy $\boldsymbol{u}_k$. We model the individual cost minimization problem as follows:

$$\min_{\boldsymbol{u}_k} \quad C_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k}) \tag{7a}$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}} x_{s,k,v} + z_{k,v} = 1, \quad \forall v \in \mathcal{V}_k, \tag{7b}$$

$$\sum_{v \in \mathcal{V}_k} c_{k,v} x_{s,k,v} + \sum_{k' \neq k} \sum_{v \in \mathcal{V}_{k'}} c_{k',v} x_{s,k',v} \leq C_s, \quad \forall s \in \mathcal{S}, \tag{7c}$$

$$x_{s,k,v}, z_{k,v} \in \{0, 1\}, \quad \forall s \in \mathcal{S}, v \in \mathcal{V}_k. \tag{7d}$$

Constraint (7b) ensures that each variant is placed on at most one ES or excluded from the edge. Constraint (7c) strictly limits the total storage consumption of all hosted variants to capacity C_s. Notably, this coupled storage constraint entwines the placement decisions of all ASPs, thereby coupling their strategy spaces.

3) GNEP Formulation: Let $\mathcal{U}_k(\boldsymbol{u}_{-k})$ denote the feasible strategy space for ASP k given rivals' strategies $\boldsymbol{u}_{-k}$, encapsulating constraints (7b)–(7d). Due to the mutual dependence in both the objective functions and the strategy spaces, the competitive interaction among all ASPs naturally constitutes a GNEP.

Game 1: (Model Variants Placement Game)

- *Players:* The set $\mathcal{K}$ of ASPs.
- *Strategies:* Each ASP $k \in \mathcal{K}$ decides its placement policy $\boldsymbol{u}_k \in \mathcal{U}_k(\boldsymbol{u}_{-k})$.
- *Objectives:* Each ASP k aims to minimize its individual operational cost $C_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k})$ in (7a).

Formally denoting this GNEP as the tuple $\mathcal{G} = \langle \mathcal{K}, \{\mathcal{U}_k(\boldsymbol{u}_{-k})\}_{k\in\mathcal{K}}, \{C_k\}_{k\in\mathcal{K}} \rangle$, a strategy profile $\boldsymbol{u}^* = (\boldsymbol{u}_1^*, \dots, \boldsymbol{u}_K^*)$ constitutes a Generalized Nash Equilibrium (GNE) for $\mathcal{G}$ if, for every ASP $k \in \mathcal{K}$, the following condition holds:

$$C_k(\boldsymbol{u}_k^*, \boldsymbol{u}_{-k}^*) \leq C_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k}^*), \quad \forall \boldsymbol{u}_k \in \mathcal{U}_k(\boldsymbol{u}_{-k}^*). \tag{8}$$

In a GNE, no ASP has the incentive to unilaterally deviate, as it cannot lower its operational cost by any deviation that remains feasible given the rivals' placements. Solving this game poses distinct theoretical and computational challenges. Unlike a standard Nash equilibrium problem (NEP), a GNEP features coupled strategy spaces, meaning the feasibility of ASP k's placement strictly depends on the residual storage left by $\boldsymbol{u}_{-k}$. Furthermore, the binary placement decisions render the feasible region non-convex and combinatorial. Furthermore, the associated social cost minimization problem is NP-hard, as shown in Appendix B. To address these issues, we next introduce an exact penalty method to decouple the storage capacity constraints. Subsequently, by exploiting the independent per-variant workload structure ($\phi_{s,k,v}$), we reformulate the penalized game as an exact potential game to facilitate equilibrium analysis.

3 Game Analysis and Distributed Algorithm Design

To address the theoretical and computational challenges of the formulated GNEP, we propose a comprehensive game-theoretic solution framework. Specifically, we first employ an exact penalty method [10] to decouple the storage capacity constraints, thereby reformulating the GNEP into a standard NEP. We then prove that this penalized game constitutes an exact potential game, enabling the design of a distributed algorithm with guaranteed finite-step convergence. We finally quantify the system's efficiency loss by deriving a parameterized Price of Anarchy (PoA) upper bound.

3.1 Game Reformulation

To eliminate the mutual dependence induced by the coupled storage constraints, we adopt an exact penalty reformulation [10]. Let $\psi_{s,k}(\boldsymbol{x}_k) \triangleq \sum_{v\in\mathcal{V}_k} c_{k,v} x_{s,k,v}$

and $\psi_s(\boldsymbol{x}) \triangleq \sum_{k\in\mathcal{K}} \psi_{s,k}(\boldsymbol{x}_k)$ denote the per-ASP and aggregate storage loads on ES s, respectively. Based on this, we design a continuous penalty function $P(\boldsymbol{x})$ to capture the system-wide risk broadcast by the ESP during resource congestion:

$$P(\boldsymbol{x}) \triangleq \eta \sum_{s\in\mathcal{S}} \max\big(0, \psi_s(\boldsymbol{x}) - C_s\big), \tag{9}$$

where η is the marginal penalty rate. Each ASP's penalized cost then becomes $\tilde{C}_k(\boldsymbol{u}) \triangleq C_k(\boldsymbol{u}) + P(\boldsymbol{x})$. To ensure strict equivalence between this penalized formulation and the original GNEP, the penalty rate η must be sufficiently large.

Lemma 2. ***(Exact Penalty Condition [10]):*** *Assuming discrete storage units where any capacity violation is at least* 1*, setting* $\eta > \max_{k,v} L_{k,v}$ *guarantees an exact penalty reformulation. Thus, any pure-strategy NE of the penalized game satisfies the original storage constraints.*

Proof. See Appendix C. ■

By absorbing the capacity coupling into the objective functions, the penalized reformulation disentangles the GNEP's interrelated feasible regions: each ASP's strategy space becomes an independent set $\mathcal{U}_k^{ind}$ governed solely by its own constraints (7b) and (7d). This transforms the original GNEP into a standard strategic game:

Game 2: (Penalized Model Variants Placement Game)

- *Players:* The set $\mathcal{K}$ of ASPs.
- *Strategies:* Each ASP $k \in \mathcal{K}$ decides its placement policy $\boldsymbol{u}_k \in \mathcal{U}_k^{ind}$.
- *Objectives:* Each ASP k aims to minimize its penalized cost $\tilde{C}_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k})$.

Given other ASPs' placement strategies $\boldsymbol{u}_{-k}$, the best response of ASP k is to find an independent strategy that minimizes its own penalized cost:

$$\mathcal{BR}_k(\boldsymbol{u}_{-k}) = \arg\min_{\boldsymbol{u}_k\in\mathcal{U}_k^{ind}} \tilde{C}_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k}). \tag{10}$$

The Nash equilibrium of this placement game is a strategy profile under which each ASP plays its best response. We next formally define the Nash equilibrium of Game 2.

Definition 1. ***(Nash Equilibrium):*** *A strategy profile* $\boldsymbol{u}^* = (\boldsymbol{u}_1^*, \dots, \boldsymbol{u}_K^*)$ *is a Nash Equilibrium (NE) of Game 2 if, for every ASP* $k \in \mathcal{K}$*:*

$$\tilde{C}_k(\boldsymbol{u}_k^*, \boldsymbol{u}_{-k}^*) \le \tilde{C}_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k}^*), \quad \forall \boldsymbol{u}_k \in \mathcal{U}_k^{ind}. \tag{11}$$

At an NE, no ASP has the incentive to unilaterally deviate from its current placement policy. Crucially, by virtue of the exact penalty condition established in Lemma 2, any standard NE of Game 2 is feasible with respect to the coupled storage constraints, thereby constituting a valid GNE for the original Game 1.

3.2 Exact Potential Game and Distributed Algorithm

We prove the interaction among ASPs constitutes an exact potential game to guarantee equilibrium existence, based on which we design a distributed algorithm.

Theorem 1. ***(Exact Potential Game):*** *Game 2 is an exact potential game with a potential function:*

$$\begin{aligned} \Pi(\boldsymbol{u}) = \frac{\beta}{2} \sum_{s \in \mathcal{S}} \left[\Phi_s(\boldsymbol{x})^2 + \sum_{k \in \mathcal{K}} \phi_{s,k}(\boldsymbol{x}_k)^2 \right] + \sum_{k \in \mathcal{K}} \sum_{v \in \mathcal{V}_k} L_{k,v}\, z_{k,v} \\ + \eta \sum_{s \in \mathcal{S}} \max\big(0, \psi_s(\boldsymbol{x}) - C_s\big), \end{aligned} \tag{12}$$

such that any unilateral deviation by ASP k from $\boldsymbol{u}_k$ to $\boldsymbol{u}'_k$ yields $\tilde{C}_k(\boldsymbol{u}'_k, \boldsymbol{u}_{-k}) - \tilde{C}_k(\boldsymbol{u}_k, \boldsymbol{u}_{-k}) = \Pi(\boldsymbol{u}'_k, \boldsymbol{u}_{-k}) - \Pi(\boldsymbol{u}_k, \boldsymbol{u}_{-k})$.

Proof. See Appendix D. ■

The exact potential game property endows Game 2 with the Finite Improvement Property (FIP) [11]. Specifically, any sequence of unilateral cost-improving updates strictly decreases the potential function $\Pi(\boldsymbol{u})$ and is guaranteed to terminate at an NE. Based on this, we design an Efficient Model Variants Placement (EMVP) algorithm (Algorithm 1).

Specifically, the algorithm initializes with a null strategy profile. In each iteration, every ASP locally computes its best-response strategy and evaluates its candidate cost reduction. The ESP then selects only the ASP with the maximum cost reduction to update its policy in each round. This unilateral update strictly decreases $\Pi(\boldsymbol{u})$. The process repeats iteratively until no ASP can further reduce its cost (i.e., $\Delta\tilde{C}_k = 0, \forall k \in \mathcal{K}$). Leveraging the FIP, this mechanism guarantees finite-step termination at a valid NE.

3.3 PoA Analysis

While equilibria guarantee system stability, ASPs' selfish nature may degrade overall efficiency. We define the social cost as the aggregate operational cost:

$$SC(\boldsymbol{u}) \triangleq \sum_{k \in \mathcal{K}} C_k(\boldsymbol{u}), \tag{13}$$

and quantify this efficiency loss via the PoA.

Definition 2. ***(PoA):*** *The Price of Anarchy [12] is the ratio of the worst-case social cost at an NE $\boldsymbol{u}^* \in NE$ to the globally optimal social cost $SC(\boldsymbol{u}^{opt})$:*

$$PoA \triangleq \max_{\boldsymbol{u}^* \in NE} \frac{SC(\boldsymbol{u}^*)}{SC(\boldsymbol{u}^{opt})}. \tag{14}$$

Algorithm 1 Efficient Model Variants Placement (EMVP) Algorithm

```
Initialization: Initialize all ASPs with the null strategy profile u^(0) (x^(0)_{s,k,v} = 0, z^(0)_{k,v} = 1). Set t ← 0.
repeat
    for each ASP k ∈ K do
        Calculate best response ũ_k by (10).
        Calculate cost reduction: ΔC̃_k = C̃_k(u_k^(t), u_{-k}^(t)) − C̃_k(ũ_k, u_{-k}^(t)).
    end for
    Selection: Identify the ASP k* = arg max_{k∈K} ΔC̃_k with the maximum improvement.
    Update: u_{k*}^(t+1) ← ũ_{k*}, and for all others u_{-k*}^(t+1) ← u_{-k*}^(t).
    t ← t + 1.
until ΔC̃_k = 0 for all k ∈ K.
Output: The NE u* = u^(t).
```

Standard PoA proofs construct Nash inequalities by assuming unilateral deviations to socially optimal strategies. Under coupled capacities, such hypothetical deviations might trigger the large exact penalty (Lemma 2), dominating the worst-case analysis and making a fixed constant bound intractable. We thus parameterize the bound using a normalized penalty ratio:

$$\rho \triangleq \frac{\eta \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} c_{k,v}}{SC(\boldsymbol{u}^{opt})}, \tag{15}$$

yielding the following guarantee.

Theorem 2. ***(General Bound under Constrained Storage):*** *Let $b = 1+\rho$ with ρ defined in* (15)*. Then*

$$PoA \leq \frac{1+2b+\sqrt{1+4b}}{2}. \tag{16}$$

Proof. See Appendix E. ■

This parameterized bound is conservative as it accounts for extreme hypothetical violations. A tighter bound emerges under abundant storage ($C_s \to \infty$). In this regime, capacity coupling vanishes, allowing $\eta = 0$ so that Game 2 reduces to a standard NEP. Setting $\eta = 0$ implies $\rho = 0$ and $b = 1$, yielding:

Lemma 3. ***(Bound under Abundant Storage):*** *If edge storage is abundant, then Game 2 is a standard NEP and*

$$PoA \leq \frac{3+\sqrt{5}}{2} \approx 2.618. \tag{17}$$

Proof. See Appendix F. ■

Lemma 3 isolates the efficiency loss purely induced by computational congestion. As verified in Section 4, actual efficiency losses in practice remain well below these theoretical worst-case limits.

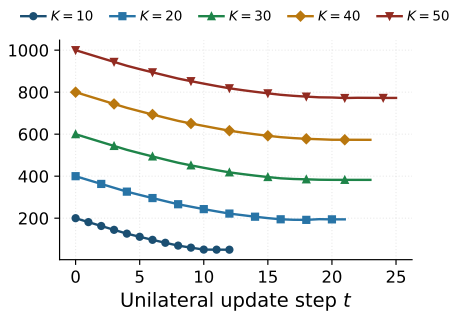

Fig. 2. Convergence of EMVP: Social cost vs. update steps across K.

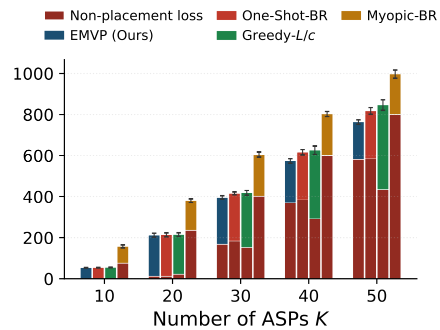

Fig. 3. Performance comparison: Social cost vs. ASP count K.

4 Simulation Results

We demonstrate the performance of the proposed EMVP algorithm compared with three benchmarks:

- **One-Shot-BR:** Each ASP updates its placement exactly once sequentially.
- **Greedy-L/c:** A heuristic prioritizing variants with the highest penalty-to-size ratio (L/c).
- **Myopic-BR:** Each ASP optimizes assuming it is the sole user of the network.

Unless stated otherwise, the default parameter settings are as follows: $V_k = V = 2$, $c_{k,v} \in \{1, 2, 3\}$, $n_{k,v} \sim \mathcal{U}(20, 50)$, $f_{k,v} \sim \mathcal{U}(1, 2)$, $S = 8$, $F_s = 1000$, $\beta = 20$, and $L_{k,v} = L = 10$, with the penalty rate set to $\eta = L + 1$ (Lemma 2). Algorithm 1 is initialized from the all-non-placed profile, and we set $C_s = 10$ in the large-scale experiments. Specifically, Figs. 2 and 3 present results averaged over five random seeds for $K \in \{10, 20, 30, 40, 50\}$; Fig. 4 fixes $K = 30$; and Fig. 5 enumerates the Nash equilibria on a small-scale instance ($K = 4$, $S = 3$) for $L \in \{5, 10, 20, 50, 100\}$.

Convergence. Fig. 2 demonstrates the rapid convergence of Algorithm 1, with the social cost stabilizing within 12–25 steps across all system sizes ($K \in \{10, \ldots, 50\}$), validating the FIP established in Theorem 1. The convergence traces exhibit a distinct two-phase pattern. Initially, ASPs freely deploy beneficial variants, driving a steep descent in social cost. However, as the number of ASPs increases, the limited edge servers cannot accommodate all variants. Consequently, the process quickly enters a constraint-dominated phase where no extra capacity is available to deploy the remaining variants, leading to rapid convergence even under large system scales.

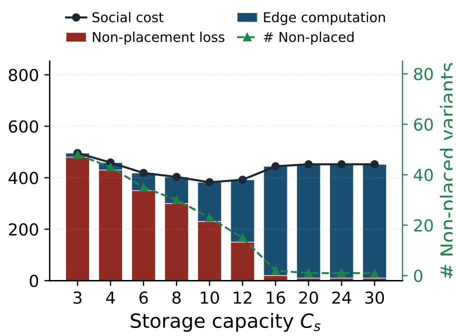

Fig. 4. Impact of capacity: Social cost and non-placed variants vs. C_s.

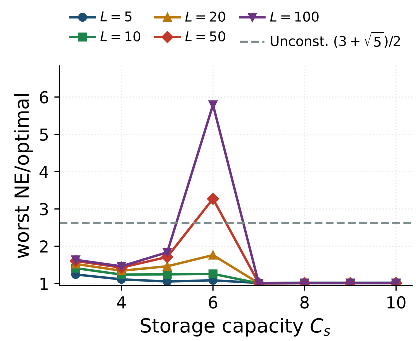

Fig. 5. Worst-NE ratio: Efficiency loss vs. capacity C_s across L.

Social cost under congestion. Fig. 3 shows that EMVP consistently achieves the lowest social cost, reducing it by 7.1%, 10.9%, and 30.6% compared to One-Shot-BR, Greedy-L/c, and Myopic-BR at extreme load ($K = 50$). The cost decomposition highlights EMVP's superior trade-off management: unlike Greedy-L/c, which triggers severe congestion by aggressively packing variants, or One-Shot-BR, which suffers from suboptimal placements without refinement, EMVP strategically balances non-placement losses with computation overhead, thereby outperforming all compared methods.

Impact of storage capacity. Fig. 4 shows a U-shaped social cost trend ($K = 30$). Under scarce storage ($C_s \leq 6$), the social cost is dominated by non-placement losses, while an optimal balance is reached at $C_s = 10$. Conversely, under abundant storage ($C_s \geq 16$), unconstrained selfish placements cause a computational tragedy of the commons: severe workload congestion inflates computation costs, driving the social cost to a suboptimal plateau.

Worst-NE ratio. Fig. 5 evaluates the worst-NE ratio to reveal the joint impact of potential loss L and storage capacity C_s. Under normal settings ($L \leq 20$), the ratio remains highly ideal and well below the unconstrained bound (Lemma 3). Severe efficiency degradation only occurs in extreme scenarios where a massive potential loss (e.g., $L = 100$) coincides with a marginally sufficient capacity ($C_s = 6$), causing a sharp spike up to 5.79. Once storage becomes abundant ($C_s \geq 7$), resource contention vanishes and the ratio safely drops to 1. Across all tested scenarios, the empirical ratios strictly satisfy the theoretical worst-case bound of Theorem 2.

5 Conclusion

In this paper, we proposed a novel variant placement framework for edge networks. We modeled resource competition among ASPs as a GNEP, which was then reformulated into an exact potential game using an exact penalty method. Leveraging the Finite Improvement Property, we designed the EMVP algorithm and established analytical PoA bounds. Experimental results demonstrate that our proposed approach consistently achieves a significantly lower social cost compared to existing benchmarks, with empirical efficiency losses remaining well below the derived theoretical limits. For future work, it would be interesting to consider a more general scenario where each variant can be deployed multiple times across the network.

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A Proof of Lemma 1

We derive the optimal solution via the KKT conditions of problem (2a). For notational brevity, let $a_{k,v} \triangleq n_{k,v} f_{k,v}/F_s$, and let the sums below run over all (k, v) with $x_{s,k,v} = 1$. The Lagrangian is

$$\mathcal{L}(\boldsymbol{w}_s, \lambda) = \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} \frac{a_{k,v}}{w_{s,k,v}} + \lambda \left(\sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} w_{s,k,v} - 1 \right), \tag{18}$$

where $\lambda \geq 0$ is the dual variable associated with the resource budget constraint. The KKT conditions are:

$$\text{(Stationarity)} \quad -\frac{x_{s,k,v}\, a_{k,v}}{w_{s,k,v}^2} + \lambda\, x_{s,k,v} = 0, \quad \forall k \in \mathcal{K}, v \in \mathcal{V}_k, \tag{19}$$

$$\text{(Primal feasibility)} \quad \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} w_{s,k,v} \leq 1, \quad w_{s,k,v} > 0 \text{ for } x_{s,k,v} = 1, \tag{20}$$

$$\text{(Dual feasibility)} \quad \lambda \geq 0, \tag{21}$$

$$\text{(Complementary slackness)} \quad \lambda \left(\textstyle\sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} w_{s,k,v} - 1 \right) = 0. \tag{22}$$

Step 1: Solving the stationarity condition. From (19), for any (k, v) with $x_{s,k,v} = 1$ we obtain

$$w_{s,k,v} = \sqrt{\frac{a_{k,v}}{\lambda}}. \tag{23}$$

Step 2: Showing the resource constraint is active. The objective $\sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v}\, a_{k,v}/w_{s,k,v}$ is strictly decreasing in each $w_{s,k,v}$ with $x_{s,k,v} = 1$. If the budget constraint were slack, i.e., $\sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} w_{s,k,v} < 1$, one could increase some $w_{s,k,v}$ while maintaining feasibility, strictly reducing the objective, a contradiction to optimality. Hence $\sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} w_{s,k,v} = 1$, which by complementary slackness (22) implies $\lambda > 0$.

Step 3: Solving for λ. Substituting (23) into the active resource constraint yields

$$\frac{1}{\sqrt{\lambda}} \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} \sqrt{a_{k,v}} = 1 \quad \Longrightarrow \quad \sqrt{\lambda} = \sum_{k\in\mathcal{K}} \sum_{v\in\mathcal{V}_k} x_{s,k,v} \sqrt{a_{k,v}}. \tag{24}$$

Step 4: Deriving the closed form. Substituting (24) back into (23) gives

$$\begin{aligned} w_{s,k,v}^* &= \frac{\sqrt{a_{k,v}}}{\sum_{k'\in\mathcal{K}} \sum_{v'\in\mathcal{V}_{k'}} x_{s,k',v'} \sqrt{a_{k',v'}}} \\ &= \frac{\sqrt{n_{k,v} f_{k,v}/F_s}}{\sum_{k'\in\mathcal{K}} \sum_{v'\in\mathcal{V}_{k'}} x_{s,k',v'} \sqrt{n_{k',v'} f_{k',v'}/F_s}} \\ &= \frac{\sqrt{n_{k,v} f_{k,v}}}{\sum_{k'\in\mathcal{K}} \sum_{v'\in\mathcal{V}_{k'}} x_{s,k',v'} \sqrt{n_{k',v'} f_{k',v'}}}, \end{aligned} \tag{25}$$

where the last equality follows since the common factor $1/\sqrt{F_s}$ cancels between the numerator and denominator. This holds for any (k, v) with $x_{s,k,v} = 1$; for $x_{s,k,v} = 0$ the indicator in (3) nullifies the expression, yielding the closed form (3).

Step 5: Uniqueness and global optimality. The objective is strictly convex in $\boldsymbol{w}_s$ on the open set $\{w_{s,k,v} > 0\}$, and the feasible region is convex. Therefore the KKT conditions are both necessary and sufficient, and the unique KKT solution $\boldsymbol{w}_s^*$ is the global minimizer. □

B NP-Hardness of the Social Cost Minimization Problem

We prove the NP-hardness by demonstrating that the classical **0-1 Knapsack Problem**, which is NP-hard, is a special case of our proposed problem.

Consider a highly simplified instance of our edge computing system with only a single edge server ($S = 1$) having storage capacity C_1. Assume the latency penalty factor is zero ($\beta = 0$), rendering the computation cost negligible. Furthermore, assume each ASP $k \in \mathcal{K}$ provides exactly one model variant ($V_k = 1$).

Let $x_k \in \{0, 1\}$ denote the placement decision for ASP k's single variant, where $x_k = 1$ indicates placement (consuming storage c_k) and $x_k = 0$ indicates non-placement (incurring potential loss L_k). Under this setting, the social cost minimization problem simplifies to:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \sum_{k=1}^{K} L_k(1 - x_k) \\ \text{s.t.} \quad & \sum_{k=1}^{K} c_k x_k \le C_1, \quad x_k \in \{0, 1\}. \end{aligned}$$

Since minimizing the penalty $\sum_k L_k(1 - x_k)$ is mathematically equivalent to maximizing the retained value $\sum_k L_k x_k$, this formulation is exactly the standard **0-1 Knapsack Problem**. Here, each ASP's variant strictly corresponds to an item with weight c_k and value L_k.

Because this highly simplified special case is already NP-hard, the general problem is also NP-hard. The general setting further introduces non-linear computation costs ($\beta > 0$), multiple variants per ASP ($V_k \ge 1$), and multiple distributed edge servers ($S > 1$), which generalizes to the Non-linear Generalized Assignment Problem. □

C Proof of Lemma 2

We prove this by contradiction. Suppose $\boldsymbol{u}^*$ is a pure-strategy NE of Game 2, but it violates the storage capacity on at least one edge server s, i.e., $\psi_s(\boldsymbol{x}^*) > C_s$. Since capacities and variant sizes are measured in discrete atomic units, the minimal positive capacity violation is at least 1 unit, meaning $\psi_s(\boldsymbol{x}^*) - C_s \ge 1$.

Because the server is over-capacitated, there exists at least one ASP k that has placed a variant $m_{k,v}$ on server s (i.e., $x^*_{s,k,v} = 1$). Consider a unilateral deviation by ASP k to withdraw this variant (i.e., changing $x_{s,k,v}$ to 0 and $z_{k,v}$ to 1).

This unilateral deviation causes the following changes to ASP k's penalized cost $\tilde{C}_k$:

1. The non-placement potential loss increases by exactly $L_{k,v}$.
2. The computation cost changes by $\Delta C^{Comp}_{k,v} \leq 0$ (since withdrawing a variant drops its edge computation latency cost to zero).
3. The storage load on server s decreases by $c_{k,v} \geq 1$. The new aggregate load is $\psi'_s = \psi_s(\boldsymbol{x}^*) - c_{k,v}$. The reduction in the penalty term is $\Delta P = \eta \max(0, \psi_s(\boldsymbol{x}^*) - C_s) - \eta \max(0, \psi'_s - C_s)$.

Because $\psi_s(\boldsymbol{x}^*) - C_s \geq 1$ and $c_{k,v} \geq 1$, the penalty reduction satisfies $\Delta P \geq \eta \cdot \min\big(c_{k,v}, \psi_s(\boldsymbol{x}^*) - C_s\big) \geq \eta \cdot 1 = \eta$.

By setting $\eta > \max_{k\in\mathcal{K}, v\in\mathcal{V}_k} L_{k,v}$, we rigorously guarantee that $\Delta P \geq \eta > L_{k,v}$. Thus, the net change in ASP k's total penalized cost is:

$$\Delta\tilde{C}_k = L_{k,v} + \Delta C^{Comp}_{k,v} - \Delta P \leq L_{k,v} + 0 - \eta < 0. \tag{26}$$

This implies that ASP k can strictly decrease its cost by unilaterally withdrawing its variant, which directly contradicts the assumption that $\boldsymbol{u}^*$ is an NE. Therefore, any NE of Game 2 must be strictly feasible (i.e., $P(\boldsymbol{x}^*) = 0$), rendering the penalized Game 2 strictly equivalent to the constrained Game 1. □

D Proof of Theorem 1

We show that the potential function (12) satisfies the exact potential condition stated in Theorem 1. Recall from (5) and (6) that ASP k's total cost is

$$C_k(\boldsymbol{u}) = \sum_{v\in\mathcal{V}_k} L_{k,v}\, z_{k,v} + \beta \sum_{v\in\mathcal{V}_k}\sum_{s\in\mathcal{S}} \phi_{s,k,v}(\boldsymbol{x}_k)\cdot \varPhi_s(\boldsymbol{x}), \tag{27}$$

where we used the fact that $x_{s,k,v} = 1 \Rightarrow z_{k,v} = 0$, so $(1 - z_{k,v})\,\phi_{s,k,v}(\boldsymbol{x}_k) = \phi_{s,k,v}(\boldsymbol{x}_k)$. Recall that $\phi_{s,k}(\boldsymbol{x}_k) \triangleq \sum_v \phi_{s,k,v}(\boldsymbol{x}_k)$, so $\varPhi_s(\boldsymbol{x}) = \sum_{k'} \phi_{s,k'}(\boldsymbol{x}_{k'})$ and the computation cost becomes $C^{Comp}_k(\boldsymbol{u}) = \beta \sum_s \phi_{s,k}(\boldsymbol{x}_k)\cdot\varPhi_s(\boldsymbol{x})$.

Potential loss part. Since $\sum_v L_{k,v} z_{k,v}$ depends only on ASP k's own placement decision $\boldsymbol{z}_k$, a unilateral deviation by ASP k changes this term by $\sum_v L_{k,v}(z'_{k,v} - z_{k,v})$, which exactly matches the corresponding change in the second term of (12) (all other ASPs' $z_{k',v}$ remain fixed).

Computation cost part. Consider a unilateral deviation of ASP k from $\boldsymbol{x}_k$ to $\boldsymbol{x}'_k$, inducing $\phi_{s,k} \to \phi'_{s,k}$ and hence $\varPhi_s \to \varPhi'_s = \varPhi_s + \Delta\phi_{s,k}$ where $\Delta\phi_{s,k} \triangleq \phi'_{s,k} - \phi_{s,k}$ (only ASP k's contribution changes; $\phi_{s,k'}$ for $k' \neq k$ remains fixed).

The change in ASP k's computation cost is

$$\begin{aligned}\Delta C_k^{Comp} &= \beta \sum_s \big[\phi'_{s,k}\,\Phi'_s - \phi_{s,k}\,\Phi_s\big] = \beta \sum_s \big[\phi'_{s,k}(\Phi_s + \Delta\phi_{s,k}) - \phi_{s,k}\,\Phi_s\big] \\ &= \beta \sum_s \big[\Delta\phi_{s,k}\,\Phi_s + \phi'_{s,k}\,\Delta\phi_{s,k}\big] = \beta \sum_s \Delta\phi_{s,k}\,\big[\Phi_s + \phi_{s,k} + \Delta\phi_{s,k}\big],\end{aligned} \tag{28}$$

where the last step uses $\phi'_{s,k} = \phi_{s,k} + \Delta\phi_{s,k}$. The change in the first term of the potential function (12) is

$$\begin{aligned}\Delta\Pi^{Comp} &= \frac{\beta}{2}\sum_s \big[(\Phi'_s)^2 - \Phi_s^2\big] + \frac{\beta}{2}\sum_s \big[(\phi'_{s,k})^2 - \phi_{s,k}^2\big] \\ &= \frac{\beta}{2}\sum_s \big[2\,\Phi_s\,\Delta\phi_{s,k} + (\Delta\phi_{s,k})^2\big] + \frac{\beta}{2}\sum_s \big[2\,\phi_{s,k}\,\Delta\phi_{s,k} + (\Delta\phi_{s,k})^2\big] \\ &= \beta \sum_s \Delta\phi_{s,k}\,\big[\Phi_s + \phi_{s,k}\big] + \beta \sum_s (\Delta\phi_{s,k})^2 \\ &= \beta \sum_s \Delta\phi_{s,k}\,\big[\Phi_s + \phi_{s,k} + \Delta\phi_{s,k}\big],\end{aligned} \tag{29}$$

which exactly matches (28).

Penalty part. The penalized cost includes the continuous storage penalty $P(\boldsymbol{x}) = \eta \sum_s \max\big(0, \psi_s(\boldsymbol{x}) - C_s\big)$ from (9). When ASP k deviates from $\boldsymbol{x}_k$ to $\boldsymbol{x}'_k$, only $\psi_{s,k}$ (and hence ψ_s) changes; let $\Delta\psi_{s,k} \triangleq \psi'_{s,k} - \psi_{s,k}$ and $\psi'_s = \psi_s + \Delta\psi_{s,k}$. The change in ASP k's penalty is

$$\Delta P_k = \eta \sum_s \big[\max(0, \psi'_s - C_s) - \max(0, \psi_s - C_s)\big], \tag{30}$$

which exactly matches the change in the third term of the potential function (12), since the penalty depends only on the aggregate load ψ_s and the other ASPs' contributions $\psi_{s,k'}$ $(k' \neq k)$ remain fixed.

Combining all three parts (potential loss, computation cost, and penalty) yields $\Delta\tilde{C}_k = \Delta\Pi$, verifying the exact potential condition of Theorem 1. Since every unilateral cost-reducing deviation strictly decreases $\Pi(\boldsymbol{u})$, and Π is bounded below with a finite strategy space, the finite improvement property holds, guaranteeing the existence of at least one pure-strategy NE. With the exact penalty parameter η set according to Lemma 2, the NE of the penalized game coincides precisely with the GNE of the original constrained game. □

E Proof of Theorem 2

We establish the PoA bound by constructing a penalty-aware Nash inequality and leveraging the Cauchy-Schwarz inequality. Let $SC(\boldsymbol{u}) = \sum_{k \in \mathcal{K}} C_k(\boldsymbol{u})$ denote the social cost.

Let $\boldsymbol{u}$ be a pure-strategy NE of Game 2. By Lemma 2, $\boldsymbol{u}$ must be strictly feasible, implying $P(\boldsymbol{u}) = 0$ and $\tilde{C}_k(\boldsymbol{u}) = C_k(\boldsymbol{u})$. According to the Nash condition, no ASP can reduce its cost by unilaterally deviating to its globally optimal strategy $\boldsymbol{u}_k^{opt}$. Therefore:

$$C_k(\boldsymbol{u}) \leq C_k(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}) + P(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}). \tag{31}$$

Summing this condition over all ASPs $k \in \mathcal{K}$ yields:

$$SC(\boldsymbol{u}) \leq SC^{Loss}(\boldsymbol{u}^{opt}) + \sum_{k \in \mathcal{K}} C_k^{Comp}(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}) + \sum_{k \in \mathcal{K}} P(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}). \tag{32}$$

Upon a unilateral deviation by ASP k to $\boldsymbol{u}_k^{opt}$, the aggregate workload on server s becomes $\Phi_s - \phi_{s,k} + \phi_{s,k}^{opt}$, which is strictly upper-bounded by $\Phi_s + \phi_{s,k}^{opt}$ (since $\phi_{s,k} \geq 0$). Thus, the computation cost sum can be expanded and bounded as:

$$\begin{aligned} \sum_{k \in \mathcal{K}} C_k^{Comp}(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}) &\leq \beta \sum_{s \in \mathcal{S}} \sum_{k \in \mathcal{K}} \phi_{s,k}^{opt}(\Phi_s + \phi_{s,k}^{opt}) \\ &= \beta \sum_{s \in \mathcal{S}} \Phi_s \Phi_s^{opt} + \beta \sum_{s \in \mathcal{S}} \sum_{k \in \mathcal{K}} (\phi_{s,k}^{opt})^2 \\ &\leq \beta \sum_{s \in \mathcal{S}} \Phi_s \Phi_s^{opt} + SC^{Comp}(\boldsymbol{u}^{opt}), \end{aligned} \tag{33}$$

where the last inequality relies on $\sum_k (\phi_{s,k}^{opt})^2 \leq (\sum_k \phi_{s,k}^{opt})^2 = (\Phi_s^{opt})^2$, and the definition $SC^{Comp}(\boldsymbol{u}^{opt}) = \beta \sum_s (\Phi_s^{opt})^2$.

Applying the Cauchy-Schwarz inequality to the cross-term in (33), we obtain:

$$\beta \sum_{s \in \mathcal{S}} \Phi_s \Phi_s^{opt} \leq \sqrt{SC^{Comp}(\boldsymbol{u}^{opt}) \cdot SC^{Comp}(\boldsymbol{u})}. \tag{34}$$

Substituting (33) and (34) back into (32), and noting that $SC^{Loss}(\boldsymbol{u}^{opt}) + SC^{Comp}(\boldsymbol{u}^{opt}) = SC(\boldsymbol{u}^{opt})$ and $SC^{Comp}(\boldsymbol{u}) \leq SC(\boldsymbol{u})$, we derive the generalized inequality:

$$SC(\boldsymbol{u}) \leq SC(\boldsymbol{u}^{opt}) + \sqrt{SC(\boldsymbol{u}^{opt}) \cdot SC(\boldsymbol{u})} + \Gamma, \tag{35}$$

where $\Gamma = \sum_k P(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k})$ is the aggregate penalty potential.

We further bound Γ by instance parameters only. Because $\boldsymbol{u}$ is feasible ($\psi_s(\boldsymbol{u}) \leq C_s$), replacing $\boldsymbol{u}_k$ by $\boldsymbol{u}_k^{opt}$ changes the load on server s by at most the storage of k's newly placed variants. Hence

$$\sum_{s \in \mathcal{S}} \max\bigl(0, \psi_s(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}) - C_s\bigr) \leq \sum_{v \in \mathcal{V}_k} c_{k,v}, \tag{36}$$

which implies $P(\boldsymbol{u}_k^{opt}, \boldsymbol{u}_{-k}) \leq \eta \sum_v c_{k,v}$ and therefore $\Gamma \leq \eta \sum_{k,v} c_{k,v}$, which is exactly the numerator of ρ defined in (15). Substituting this uniform upper bound into (35) yields

$$SC(\boldsymbol{u}) \leq SC(\boldsymbol{u}^{opt}) + \sqrt{SC(\boldsymbol{u}^{opt}) \cdot SC(\boldsymbol{u})} + \eta \sum_{k,v} c_{k,v}. \tag{37}$$

Dividing both sides by $SC(\boldsymbol{u}^{opt})$ and defining the PoA ratio $\alpha = \frac{SC(\boldsymbol{u})}{SC(\boldsymbol{u}^{opt})}$, we have:

$$\alpha \leq 1 + \sqrt{\alpha} + \frac{\eta \sum_{k,v} c_{k,v}}{SC(\boldsymbol{u}^{opt})}. \tag{38}$$

Recalling ρ from (15) and letting $b = 1 + \rho$ as in Theorem 2, the inequality simplifies to $\alpha - \sqrt{\alpha} - b \leq 0$. Substituting $z = \sqrt{\alpha}$, the positive root of the quadratic inequality $z^2 - z - b \leq 0$ yields:

$$z \leq \frac{1 + \sqrt{1 + 4b}}{2}. \tag{39}$$

Squaring z gives the closed-form PoA bound under constrained storage:

$$\text{PoA} = \alpha = z^2 \leq \frac{1 + 2b + \sqrt{1 + 4b}}{2}. \tag{40}$$

□

F Proof of Lemma 3

We build on the derivation in Appendix E. When edge storage is abundant ($C_s \to \infty$, $\forall s \in \mathcal{S}$), the storage constraints never bind, so the exact penalty is unnecessary and we can set $\eta = 0$. In this regime, no unilateral deviation to $\boldsymbol{u}_k^{opt}$ can violate any capacity, the aggregate hypothetical penalty vanishes ($\Gamma = 0$), and the penalized cost $\widetilde{C}_k$ coincides with C_k, i.e., Game 2 reduces to a standard NEP. The inequality (38) therefore reduces to the case $b = 1$ (equivalently, $\rho = 0$ in (15)). Substituting $b = 1$ into (40) yields

$$\text{PoA} \leq \frac{1 + 2(1) + \sqrt{1 + 4(1)}}{2} = \frac{3 + \sqrt{5}}{2}. \tag{41}$$

□